

1. Which number(s) is(are) equal to its (their) square?

Answer. 0, 1

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2. Which number(s) is(are) equal to half its (their) square?

Answer. 0, 2

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3. Which number(s) is(are) equal to the quarter of its (their) square?

Answer. 0, 4

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4. A car travels from A to B at a speed of 40 mph then returns, using the same road, from B to A at a speed of 60 mph. What is the average speed for the round trip?

Answer. 48 miles per hour

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5. Tom travels 60 miles per hour going to a neighboring city and 50 miles per hour coming back using the same road. He drove a total of 5 hours away and back. What is the distance from Tom's house to the city he visited? (round your answer to the nearest mile).

Answer. 136 miles

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6. At 11:00 a.m., John started driving along a highway at constant speed of 50 miles per hour. A quarter of an hour later, Jimmy started driving along

the same highway in the same direction as John at the constant speed of 65 miles per hour. At what time will Jimmy catch up with John?

Answer. 12:05 a.m.

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7. Find an equation of the line containing  $(-4, 5)$  and perpendicular to the line  $5x - 3y = 4$ .

Answer.  $5y + 3x = 13$

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8. A rectangle field has an area of 300 square meters and a perimeter of 80 meters. What are the length and width of the field?

Answer. length = 30 meters , width = 10 meters

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9. Find the area of a trapezoid whose parallel sides are 12 and 23 centimeters respectively.

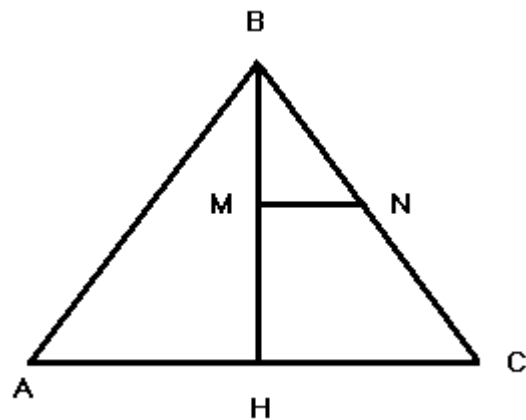
Answer. not enough information to solve the problem.

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10. A rectangular garden in Mrs Dorothy's house has a length of 100 meters and a width of 50 meters. A square swimming pool is to be constructed inside the garden. Find the length of one side of the swimming pool if the remaining area (not occupied by the pool) is equal to one half the area of the rectangular garden.

Answer. 50 meters

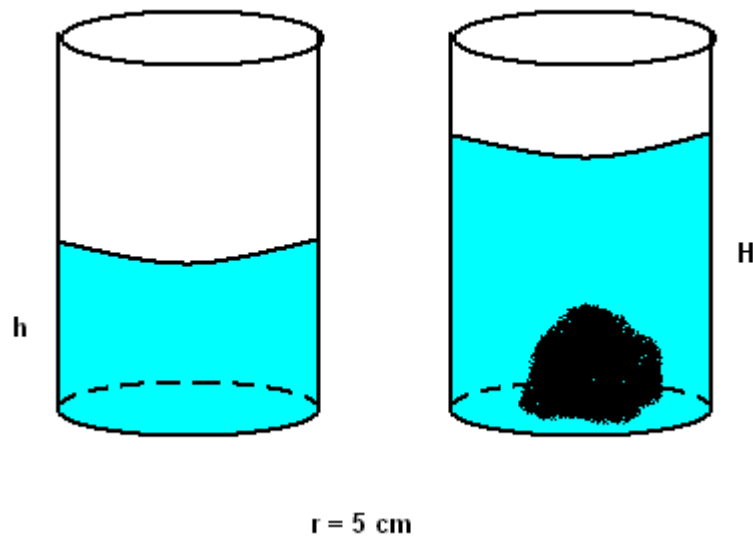
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11. ABC is an equilateral triangle with side length equal to 50 cm. BH is perpendicular to AC. MN is parallel to AC. Find the area of triangle BMN if the length of MN is equal to 12 cm.



Answer.  $72\sqrt{3}$  square centimeters

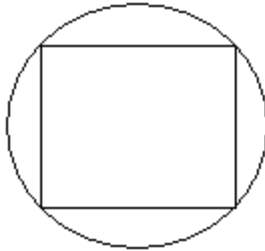
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12. The height  $h$  of water in a cylindrical container with radius  $r = 5$  cm is equal to 10 cm. Peter needs to measure the volume of a stone with a complicated shape and so he puts the stone inside the container with water. The height of the water inside the container rises to 13.2 cm. What is the volume of the stone in cubic cm?



Answer. 80 pi cubic centimeters

13. In the figure below the square has all its vertices on the circle. The area of the square is equal to 400 square cm. What is the area of the circular shape?



Answer. 200 pi square centimeters

14. The numbers 2 , 3 , 5 and x have an average equal to 4. What is x?

Answer.  $x = 6$

15. The numbers  $x$ ,  $y$ ,  $z$  and  $w$  have an average equal to 25. The average of  $x$ ,  $y$  and  $z$  is equal to 27. Find  $w$ .

Answer.  $w = 19$

16. Find  $x$ ,  $y$ ,  $z$  so that the numbers 41, 46,  $x$ ,  $y$ ,  $z$  have a mean of 50 and a mode of 45.

Answer.  $x = 45$ ,  $y = 45$  and  $z = 73$

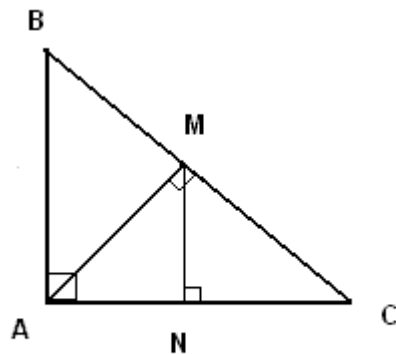
17.  $A$  is a constant. Find  $A$  such that the equation  $2x + 1 = 2A + 3(x + A)$  has a solution at  $x = 2$ .

Answer.  $A = -1/5$

18. 1 liter is equal to 1 cubic decimeter and 1 liter of water weighs 1 kilogram. What is the weight of water contained in a cylindrical container with radius equal to 50 centimeters and height equal to 1 meter?

Answer. 250 pi kilograms

19. In the figure below triangle  $ABC$  is an isosceles right triangle.  $AM$  is perpendicular to  $BC$  and  $MN$  is perpendicular to  $AC$ . Find the ratio of the area of triangle  $MNC$  to the area of triangle  $ABC$ .



Answer. 1:4

20. Pump A can fill a tank of water in 4 hours. Pump B can fill the same tank in 6 hours. Both pumps are started at 8:00 a.m. to fill the same empty tank. An hour later, pump B breaks down and took one hour to repair and was restarted again. When will the tank be full? (round your answer to the nearest minute).

Answer. 10:48 a.m.

21. Are the lines with equations  $2x + y = 2$  and  $x - 2y = 0$  parallel, perpendicular or neither?

Answer. perpendicular

22. What are the dimensions of the square that has the perimeter and the area equal in value?

Answer.

23. Find the dimensions of the rectangle that has a length 3 meters more than its width and a perimeter equal in value to its area?

Answer. square with side length = 4 units

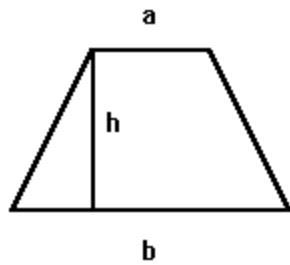
24. Find the circumference of a circular disk whose area is  $100\pi$  square centimeters.

Answer. length = 6 units and width = 3 units

25. The semicircle of area  $50\pi$  centimeters is inscribed inside a rectangle. The diameter of the semicircle coincides with the length of the rectangle. Find the area of the rectangle.

26.  $20\pi$  centimeters

4)



The area of a trapezoid is given by the formula  $A = \frac{1}{2} (a + b) h$ . If  $h = \frac{1}{2}b$  and  $a = \frac{1}{3}b$ .

a) What will be the value of  $b$  when  $a = 26$  ?

b) What will be the value of  $h$  if  $a = 6$ ? What will be the area of the trapezoid?

Answer:

$$A) \quad 26 = \frac{1}{2} \left( \frac{4}{3} b \right) \left( \frac{1}{2} b \right)$$

$$26 = \frac{4}{12} b$$

$$26(12)/4 = b$$

$$b = 78$$

$$B) \quad \frac{1}{3}b = a \quad a = \frac{1}{2} (6+18) (9)$$

$$b = 3(6) = 18 \quad = 108$$

$$h = (1/2b) = 9$$

5) Two girls agree to go on a road trip together. They travel  $(x + 5)$  km on the first day. On the second day they travel 2 km more than half of the distance they travelled on the first day. On the third day they drove 3 times as far as they did on the second day. If they drove 5000 km total, find the value of  $x$ .

Answer:

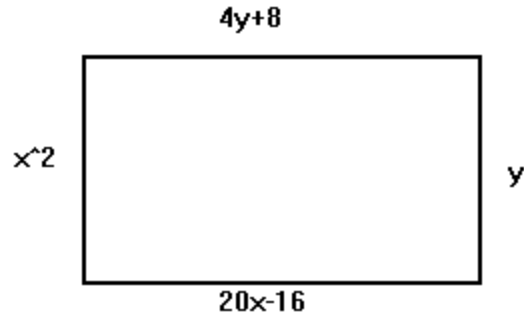
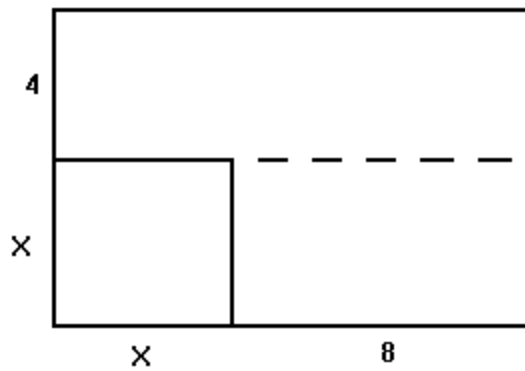
$$(x+5) + \frac{1}{2}(x+5) + 2 + 3 \left[ \frac{1}{2}(x+5) + 2 \right] = 5000$$

$$x+5 + \frac{1}{2}x + 45 + \frac{3}{2}x + 7.5 + 6 = 5000$$

$$3x + 23 = 5000$$

$$x = \frac{5000-23}{3} = 1659$$

6) A rectangle has the following sides. One side must have an odd integer length. Find the perimeter.



**Answer:**

We know that multiplication by an even number makes an odd number even

so  $y+8$  and  $20x-16$  are even we need  $y^2$  and  $x^2$  to be odd.

$$y=x^2$$

$$4y+8 = 20x-16$$

$$4x^2 - 20x + 24 = 0$$

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0$$

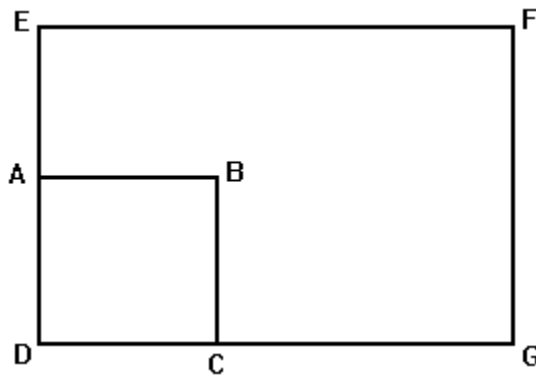
$$x = 2, x = 3$$

$$x^2 = 4 \text{ or } x^2 = 9$$

$$\text{Therefore } y = 9$$

$$\begin{aligned} \text{Perimeter} &= x^2 + 4y + 8 + y + 20x - 16 \\ &= 9 + 36 + 8 + 9 + 60 - 16 \\ &= 106 \end{aligned}$$

7) From the rectangle DEFG the square ABCD is removed, leaving an area of 92 square units. If AE is 4 and GC is 8 units, then use a polynomial expression to find the original area of DEFG.



**Answer:**

$$\begin{aligned} 92 &= 4(x+8) + 8(x) = 12x + 32 = 92 \\ 12x &= 60 \end{aligned}$$

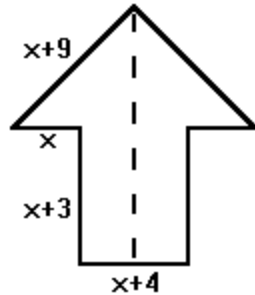


$$x = 5$$

$$(x+8)(x+4) = A(DEFG)$$

$$(13)(9) = 117$$

8) Given the symmetric shape below with a known perimeter of 77, find the area of the shape.



Answer:

$$2(x+9+x+x+3)+x+4 = 77$$

$$2x+18 + 2x + 2x +6 + x+4 = 77$$

$$7x+28 = 77$$

$$7x = 49$$

$$x = 7$$

$$\text{Therefore } x+9 = 16$$

$$x = 7$$

$$x+3 = 10$$

$$x+4 = 11$$

$$16^2 = 7^2 + h^2 \quad h=14.39$$

$$A = 1/2 (2x+x+4) (14.39)$$

$$= 179.84$$

$$A_{\text{sect}} + (x+3)(x+4) = (10)(11) = 110$$

$$\text{Therefore } A_{\text{total}} = 179.84 + 110 = 289.84$$

9) Find an odd number with 3 digits such that all the digits are different and add up to 15. The difference between the first two digits equals the difference between the last two digits. The hundreds digit is greater than the sum of the tens and ones digits.

Answer:

Digit ABC

$$\text{Such that } a+b+c = 15$$

$$\text{and } a-b = b-c$$

$$a = 2b-c$$

$$2b+b = 15$$

$$b = 5$$

$$a+c = 10$$

$$a = 10-c$$

$$a > b+c$$

$$10-c > b+c$$

$$10-b > 2c$$

$$5 > 2c$$

$$2.5 > c$$

c must be odd, therefore  $c=1$ ,  $b=5$ ,  $c=1$ ,  $a=9$

10) A real estate agent received a 6% commission on the selling price of a house. If his commission was \$8,880, what was the selling price of the house?

Answer

$x = \$148,000$

11) An electric motor makes 3,000 revolutions per minutes. How many degrees does it rotate in one second?

- A. Answer: 3000 revolutions / minute  
=  $3000 \times 360$  degrees / 60 seconds  
= 18,000 degrees / second

12) The area of a rectangular field is equal to 300 square meters. Its perimeter is equal to 70 meters. Find the length and width of this rectangle.

- A.  $L \times W = 300$  : area, L is the length and W is the width.
- B.  $2L + 2W = 70$  : perimeter
- C.  $L = 35 - w$  : solve for L
- D.  $(35 - W) \times W = 300$  : substitute in the area equation
- E.  $W = 15$  and  $L = 20$  : solve for W and find L using  $L = 35 - w$ .

13) If a tire rotates at 400 revolutions per minute when the car is travelling 72km/h, what is the circumference of the tire?

- A. 400 rev / minute =  $400 \times 60$  rev / 60 minutes  
= 24,000 rev / hour
- B.  $24,000 \times C = 72,000$  m : C is the circumference
- C.  $C = 3$  meters

14) In a shop, the cost of 4 shirts, 4 pairs of trousers and 2 hats is \$560. The cost of 9 shirts, 9 pairs of trousers and 6 hats is \$1,290. What is the total cost of 1 shirt, 1 pair of trousers and 1 hat?

- A. Let x be the price of one shirt, y be the price of one pair of trousers and z be the price of one hat.
- B.  $4x + 4y + 2z = 560$  :
- C.  $9x + 9y + 6z = 1,290$
- D.  $3x + 3y + 2z = 430$  : divide all terms of equation C by 3
- E.  $x + y = 130$  : subtract equation D from equation B
- F.  $3(x + y) + 2z = 430$  : equation D with factored terms.
- G.  $3 \times 130 + 2z = 430$
- H.  $z = 20$  : solve for z
- I.  $x + y + z = 130 + 20 = \$150$

15) Four children have small toys. The first child has 1/10 of the toys, the second child has 12 more toys than the first, the third child has one more toy of what the first child has and the fourth child has double the third child. How many toys are there?

Answer.

- A.  $x$  : the total number of toys
- B.  $x/10$  : the number of toys for first child
- C.  $x/10 + 12$  : the number of toys for second child
- D.  $x/10 + 1$  : the number of toys for the third child
- E.  $2(x/10 + 1)$  : the number of toys for the fourth child
- F.  $x/10 + x/10 + 12 + x/10 + 1 + 2(x/10 + 1) = x$
- G.  $x = 30$  toys : solve for  $x$

16) A class average mark in an exam is 70. The average of students who scored below 60 is 50. The average of students who scored 60 or more is 75. If the total number of students in this class is 20, how many students scored below 60?

Answer.

- A. Let  $n$  the number of students who scored below 60 and  $N$  the number of students who scored 60 or more.  $X_i$  the grades below 60 and  $Y_i$  the grades 60 or above.
- B.  $[\text{sum}(X_i) + \text{sum}(Y_i)] / 20 = 70$  : class average
- C.  $\text{sum}(X_i) / n = 50$  : average for less than 60
- D.  $\text{sum}(Y_i) / N = 75$  : average for 60 or more
- E.  $50n + 75N = 1400$  : combine the above equations
- F.  $n + N = 20$  : total number of students
- G.  $n = 4$  and  $N = 16$  : solve the above system

17) It takes a boat 3 hours to travel down a river from point A to point B, and 5 hours to travel up the river from B to A. How long would it take the same boat to go from A to B in still water?

Answer.

- A. Let:  $S$  be the speed of the boat in still water,  $r$  be the rate of the water current and  $d$  the distance between A and B.
- B.  $d = 3(S + r)$  : boat traveling down river
- C.  $d = 5(S - r)$  : boat traveling up river
- D.  $3(S + r) = 5(S - r)$
- E.  $r = S / 4$  : solve above equation for  $r$
- F.  $d = 3(S + S/4)$  : substitute  $r$  by  $S/4$  in equation B
- G.  $d / S = 3.75$  hours = 3 hours and 45 minutes.

18) An airplane flies against the wind from A to B in 8 hours. The same airplane returns from B to A, in the same direction as the wind, in 7 hours. Find the ratio of the speed of the airplane (in still air) to the speed of the wind.

Answer.

- A. Let:  $S$  be the speed of the airplane in still air,  $r$  be the speed of the wind and  $d$  the distance between A and B.
- B.  $d = 8(S - r)$  : airplane flies against the wind
- C.  $d = 7(S + r)$  : airplane flies in the same direction as the wind
- D.  $8(S - r) = 7(S + r)$
- E.  $S/r = 15$